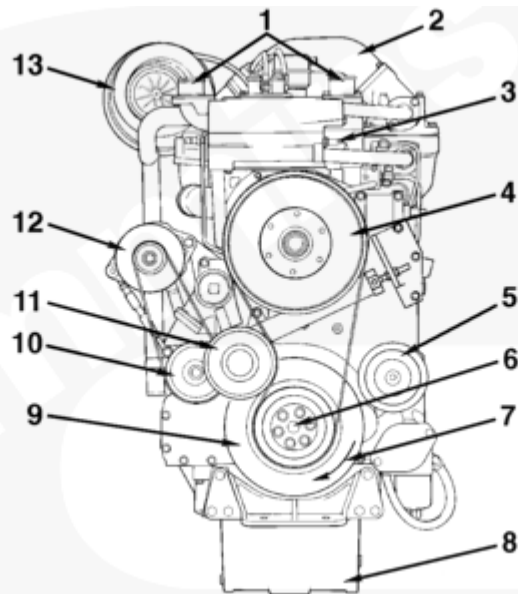


Trainings ▾

Engine Views

with Mechanically Actuated Injector

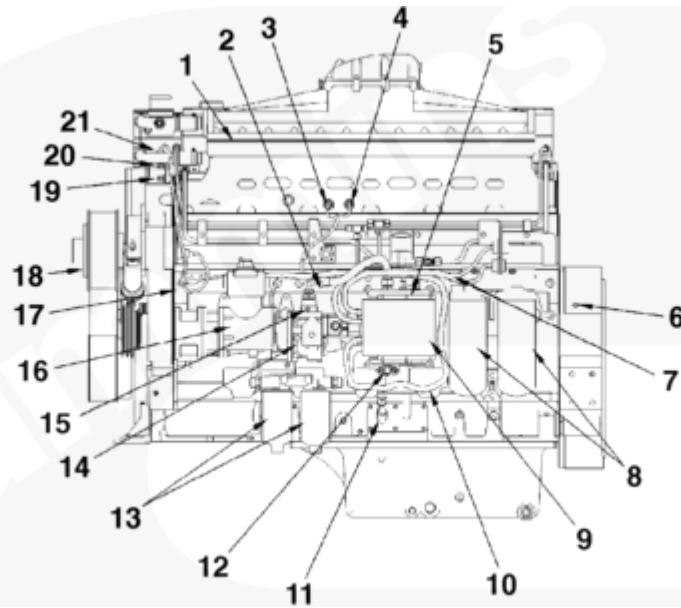
©Cummins Inc



00400258

Front View - QSK19

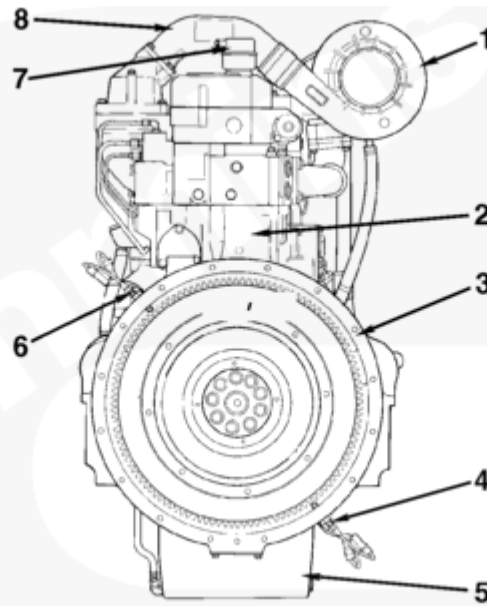
1. Coolant outlet
2. Air crossover
3. Coolant temperature sensor
4. Fan pulley
5. Accessory drive pulley
6. Crankshaft pulley
7. Direction of rotation
8. Oil pan
9. Vibration damper
10. Alternator drive pulley
11. Fan tensioner idler
12. Alternator
13. Turbocharger.



00400003

Left View - QSK19

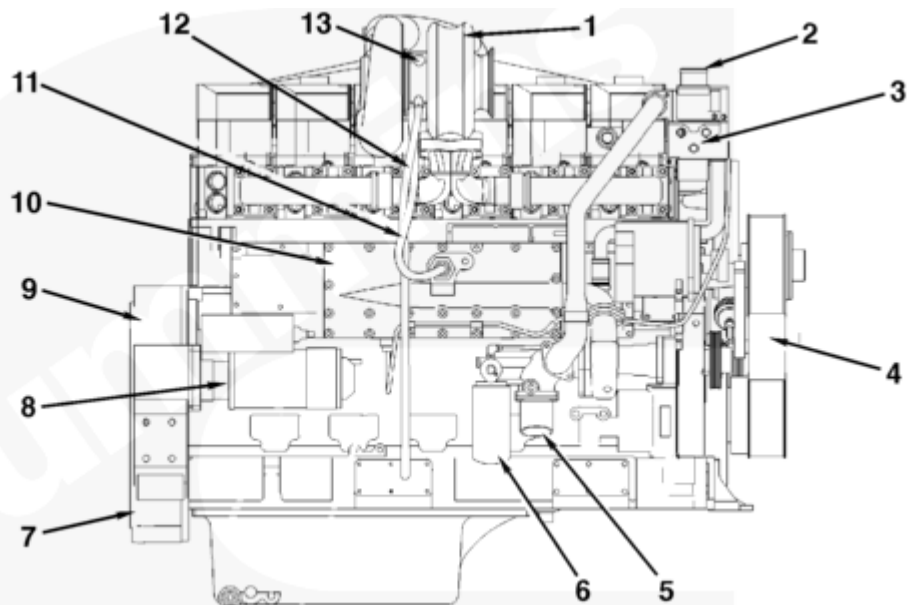
1. Aftercooler assembly
2. Engine oil rifle pressure 1 sensor
3. Intake manifold 1 temperature sensor
4. Intake manifold 1 pressure sensor
5. Control valve body
6. Engine crankshaft speed/position sensor (generator drive)
7. Original equipment manufacturer (OEM) interface harness
8. Combination full flow and bypass oil filters
9. Engine control module (ECM)
10. Engine harness
11. Dipstick
12. Barometric pressure sensor
13. Fuel filters
14. Fuel pump
15. Fuel pump outlet quick-disconnect fitting
16. Air compressor
17. Engine camshaft speed/position sensor (**not** generator drive)
18. Fan hub
19. Fuel rail quick-disconnect fitting
20. Timing rail quick-disconnect fitting
21. Engine coolant temperature 1 sensor.



00400259

Rear View - QSK19

1. Turbocharger
2. Cylinder block
3. Flywheel housing
4. Engine crankshaft speed/position sensor (**not** generator drive)
5. Oil pan
6. Engine speed/position sensor
7. Crankcase breather
8. Air crossover.

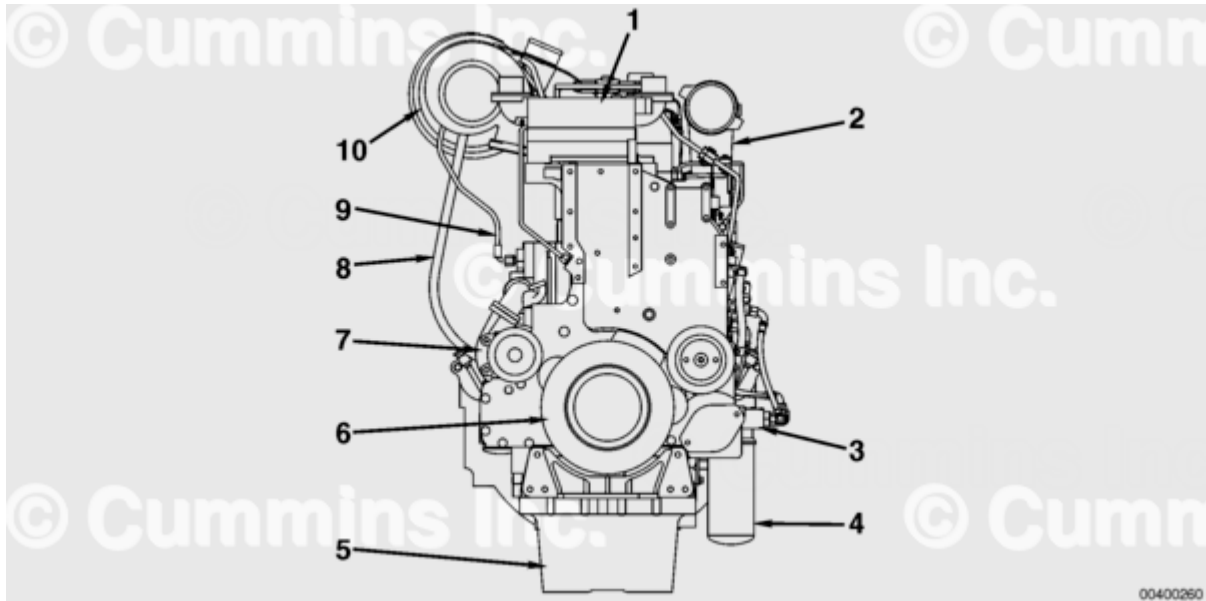


19802407

Right View - QSK19

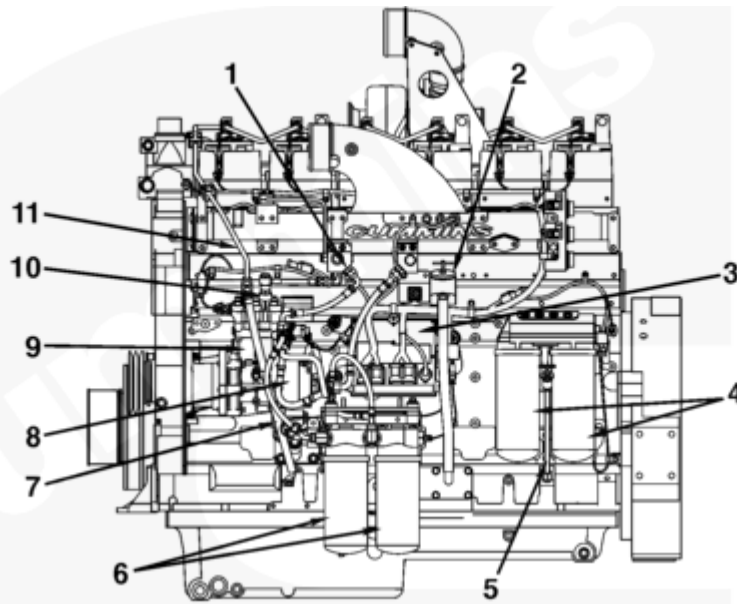
1. Turbocharger
2. Coolant outlet
3. Thermostat housing
4. Fan idler and assembly

5. Coolant inlet
6. Coolant filter
7. Alternate location for crankshaft speed/position sensor (generator drive **only**)
8. Starting motor
9. Flywheel housing
10. Oil cooler
11. Coolant supply to turbocharger
12. Turbocharger oil drain
13. Oil inlet to turbocharger.



Front View - QSK19 Industrial

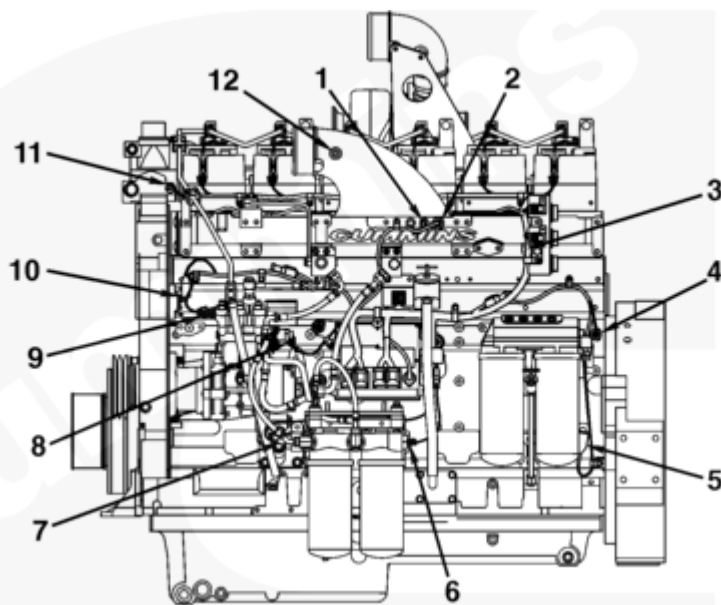
1. Thermostat housing
2. Air intake connection
3. Fuel filter head
4. Fuel filter
5. Oil pan
6. Vibration damper
7. Water pump
8. Turbocharger oil drain
9. Turbocharger oil supply
10. Turbocharger.



00400205

Left View - QSK19 Industrial

1. Five bar check valve
2. Oil fill location
3. ECM
4. Oil filters
5. Dipstick
6. Fuel filter 2nd stage
7. Double-wall injector supply vent line
8. Gerotor pump
9. Fuel pump
10. Mechanical dump valve
11. Double-wall injector supply line.

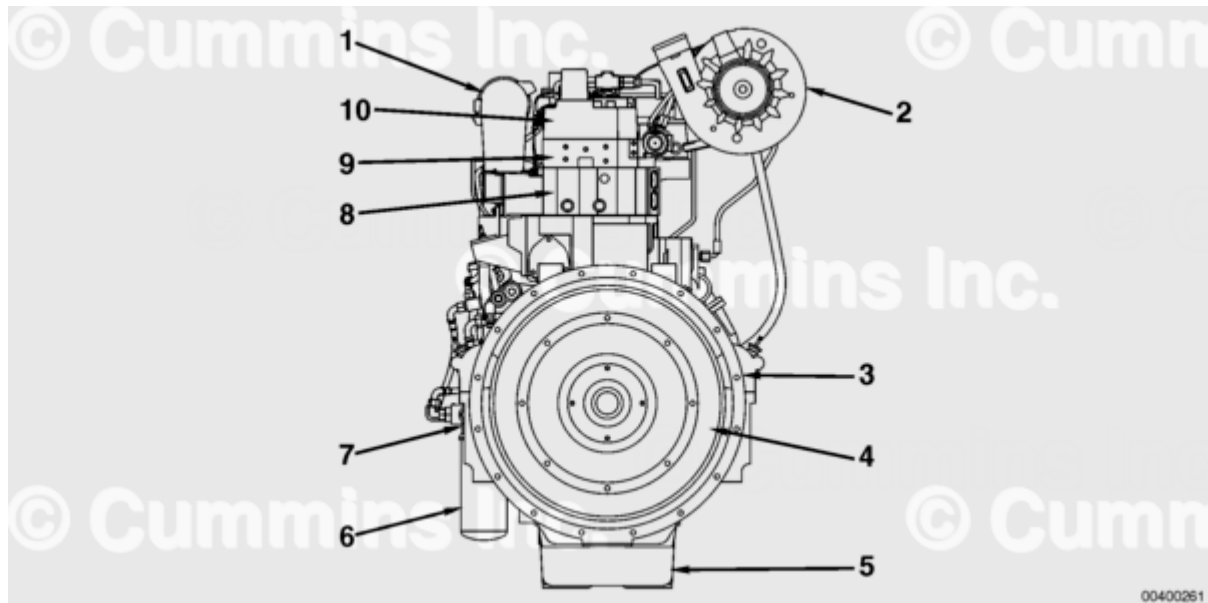


00400205

Left View - QSK19 Industrial (Sensor Locations)

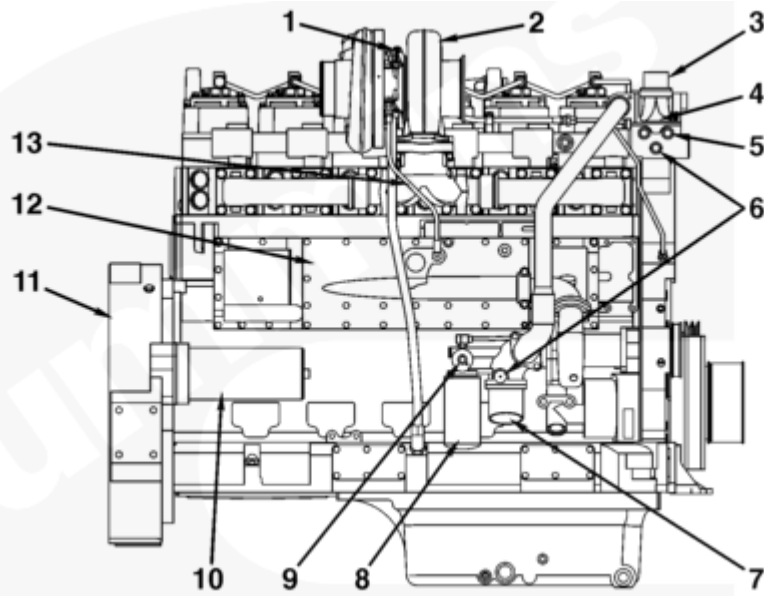
1. Intake manifold 1 temperature sensor

2. Intake manifold 1 pressure sensor
3. Barometric pressure sensor
4. Engine oil rifle pressure 1 sensor
5. Engine crankshaft speed/position sensor
6. Fuel delivery pressure sensor
7. Engine fuel temperature sensor 1
8. Fuel pump pressurizing assembly 1
9. Injector metering rail 1 pressure sensor
10. Engine camshaft speed/position sensor
11. Engine coolant temperature 1 sensor
12. Intake manifold port.



Rear View - QSK19 Industrial

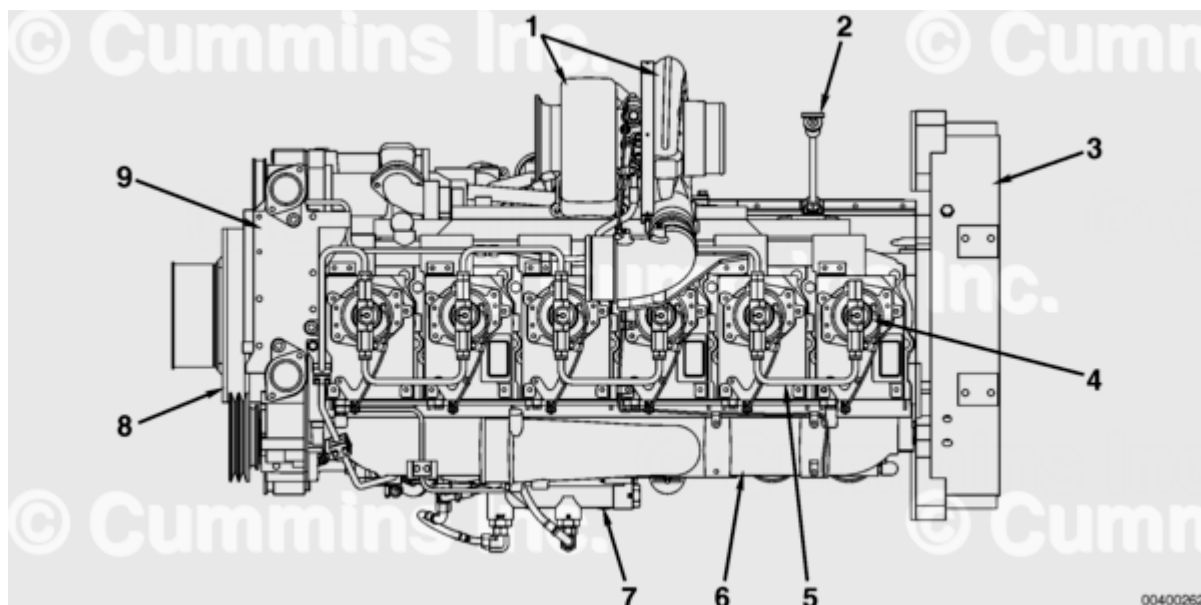
1. Air crossover connection
2. Turbocharger
3. Flywheel housing
4. Flywheel
5. Oil pan
6. Fuel filter
7. Fuel filter head
8. Cylinder head
9. Rocker lever housing
10. Valve cover.



00400204

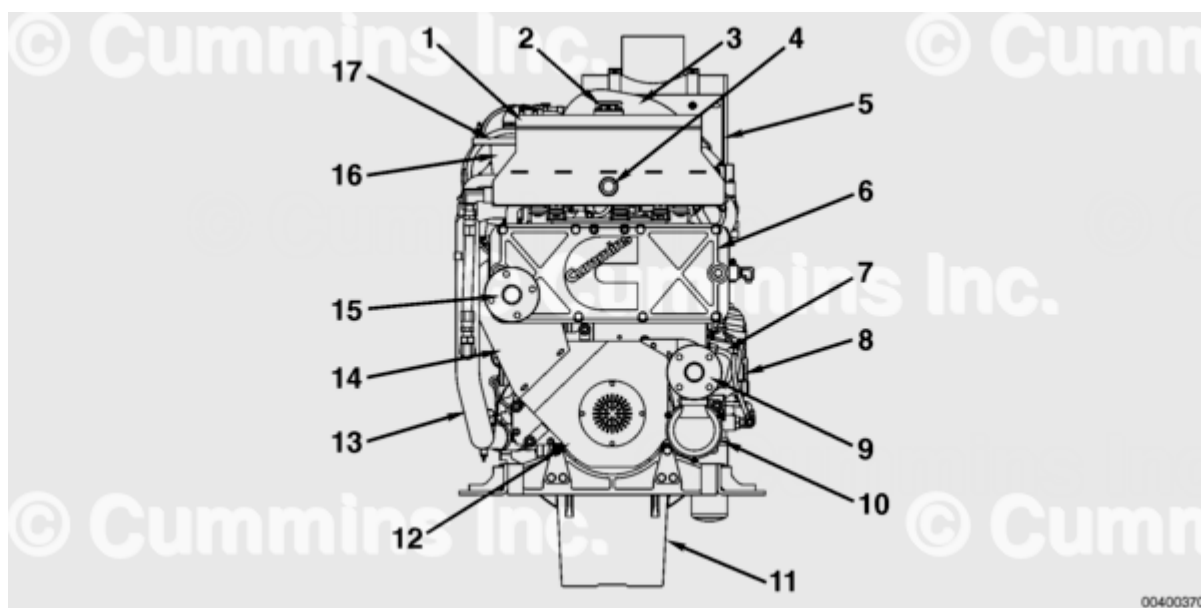
Right View - QSK19 Industrial

1. Turbocharger 1 speed sensor
2. Turbocharger
3. Coolant outlet
4. Thermostat housing
5. Coolant pressure port
6. Coolant IN and OUT restriction ports
7. Coolant inlet
8. Coolant filter
9. Coolant filter shutoff valve
10. Starting motor
11. Flywheel housing
12. Oil cooler
13. Coolant supply to turbocharger.



Top View - QSK19 Industrial

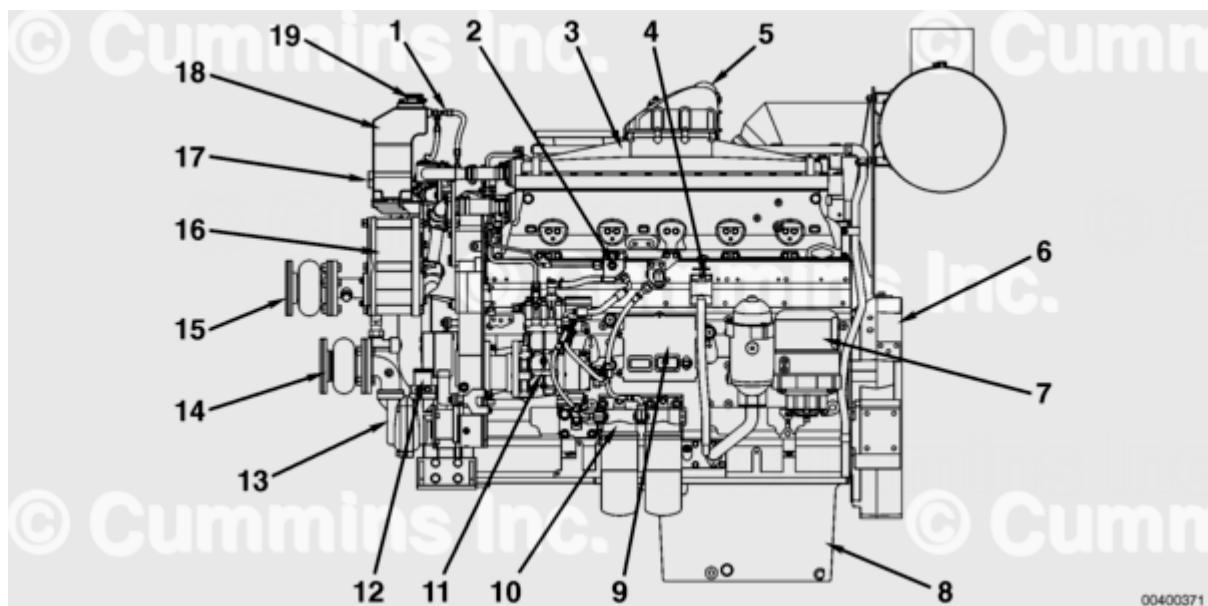
1. Turbocharger
2. Oil dipstick
3. Flywheel housing
4. Injector
5. Fuel injector supply line
6. Intake manifold
7. Fuel filter head
8. Vibration damper
9. Thermostat housing.



Front View - QSK19 CM850 Heat Exchanger

1. Expansion tank
2. Expansion tank pressure cap
3. Air crossover pipe

4. Expansion tank sight glass
5. Aftercooler
6. Heat exchanger
7. Raw water outlet to heat exchanger
8. Lubricating oil dipstick
9. Raw water inlet to raw water pump
10. Raw water pump
11. Lubricating oil pan
12. Vibration damper guard
13. Engine water pump inlet pipe
14. Alternator belt guard
15. Raw water outlet from heat exchanger
16. Exhaust elbow
17. Exhaust outlet.

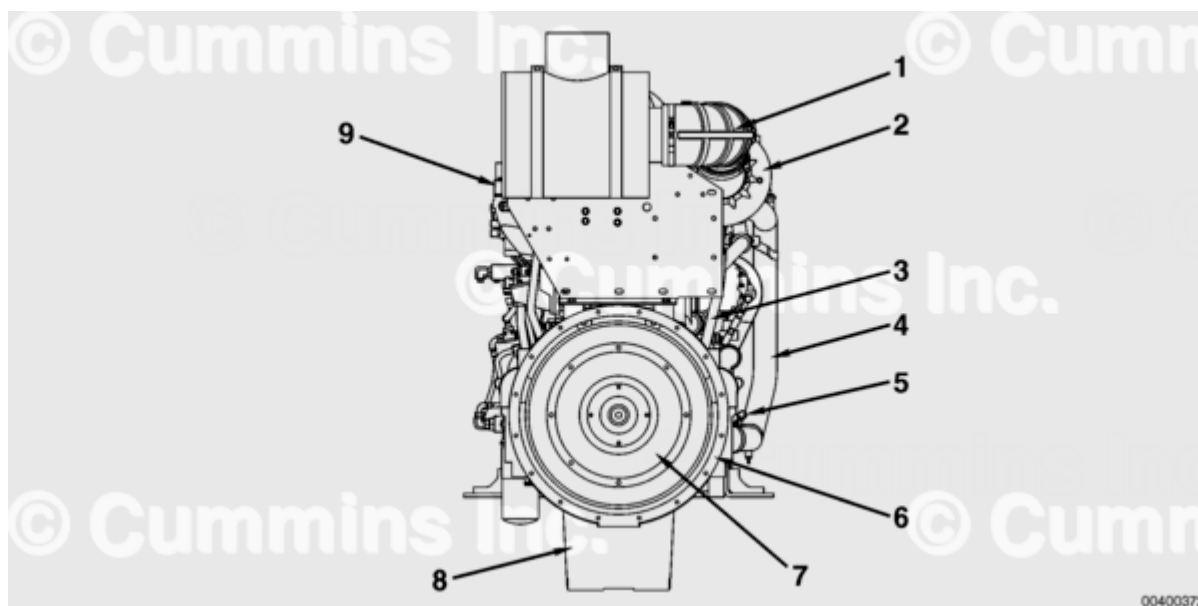


Left View - QSK19 CM850 Heat Exchanger

1. Coolant vents
2. Fuel drain manifold
3. Aftercooler
4. Oil fill location
5. Air crossover pipe
6. Flywheel housing
7. Eliminator™*
8. Lubricating oil pan
9. ECM
10. Fuel filter head
11. Fuel pump
12. Raw water pump outlet to heat exchanger
13. Raw water pump
14. Raw water pump inlet

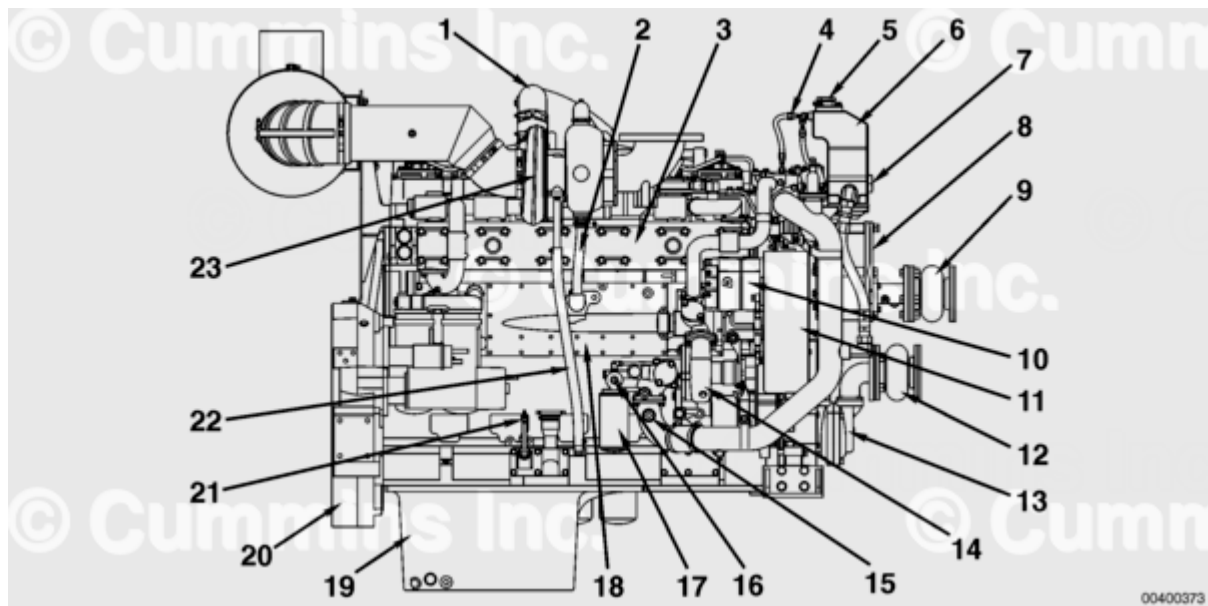
- 15. Raw water outlet from heat exchanger
- 16. Heat exchanger
- 17. Expansion tank sight glass
- 18. Expansion tank
- 19. Expansion tank cap.

* The Eliminator™ is an available option on QSK19 Modular Common Rail System Marine engines.



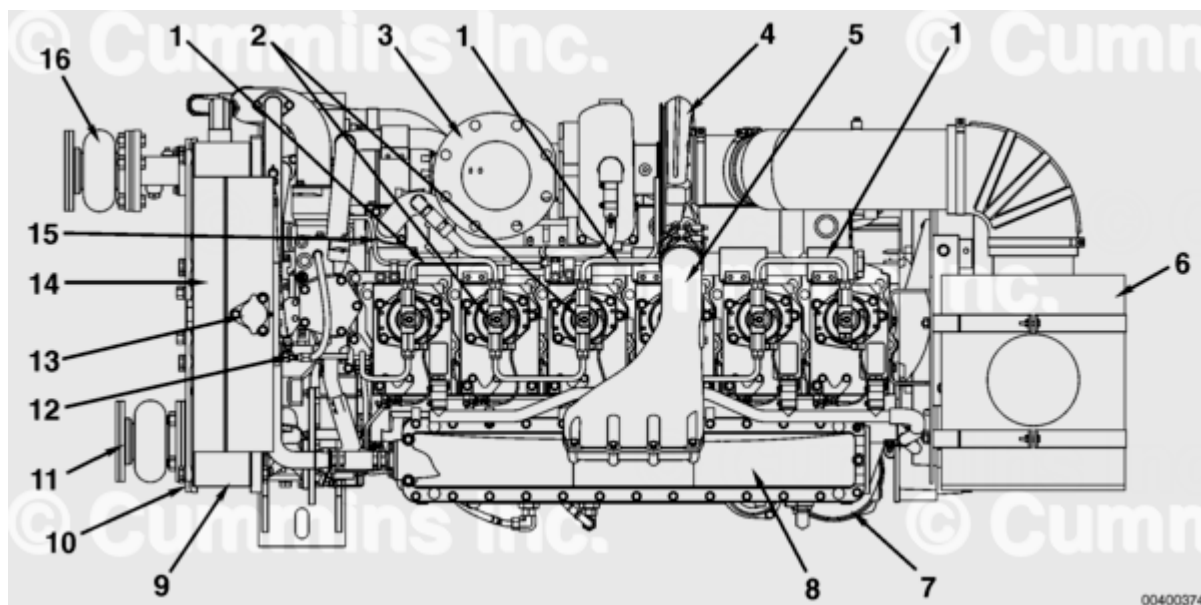
Rear View - QSK19 CM850 Heat Exchanger

- 1. Air inlet hose
- 2. Turbocharger
- 3. Turbocharger oil drain
- 4. Engine water pump inlet pipe
- 5. Lubricating oil dipstick
- 6. Flywheel housing
- 7. Flywheel
- 8. Lubricating oil pan
- 9. Aftercooler.



Right View - QSK19 CM850 Heat Exchanger

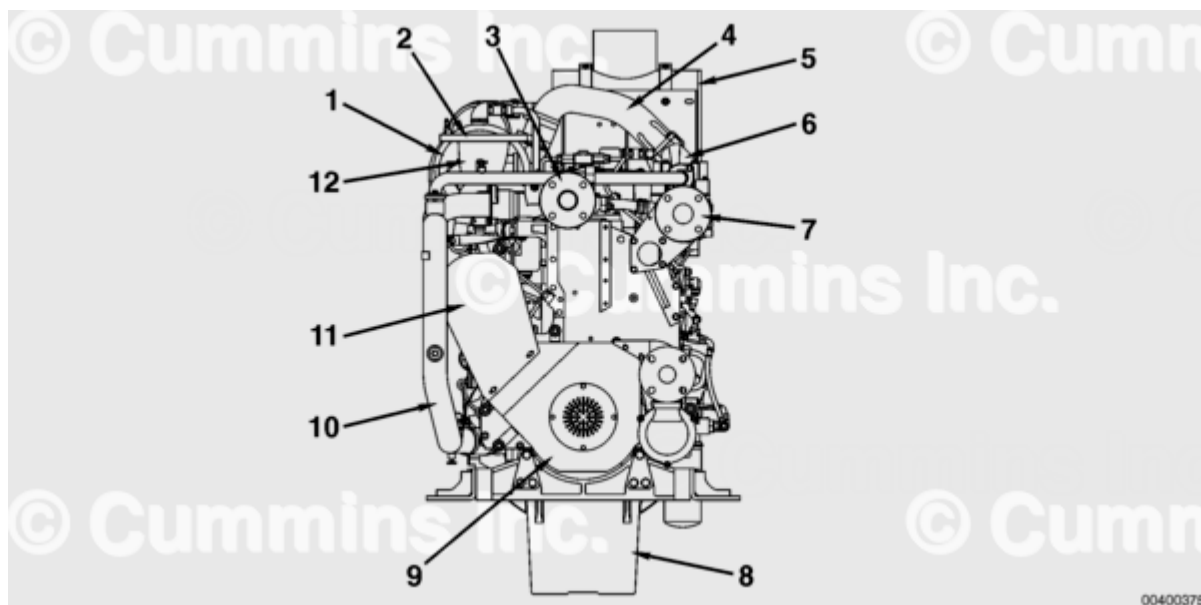
1. Air crossover pipe
2. Turbocharger coolant supply
3. Exhaust manifold
4. Coolant vent lines
5. Expansion tank pressure cap
6. Expansion tank
7. Expansion tank sight glass
8. Heat exchanger
9. Raw water outlet from heat exchanger
10. Alternator
11. Belt guard
12. Raw water inlet
13. Raw water pump
14. Water pump
15. Water inlet restriction test port
16. Water filter shutoff valve
17. Water filter
18. Lubricating oil cooler
19. Lubricating oil pan
20. Flywheel housing
21. Lubricating oil dipstick
22. Turbocharger oil drain
23. Turbocharger.



Top View - QSK19 CM850 Heat Exchanger

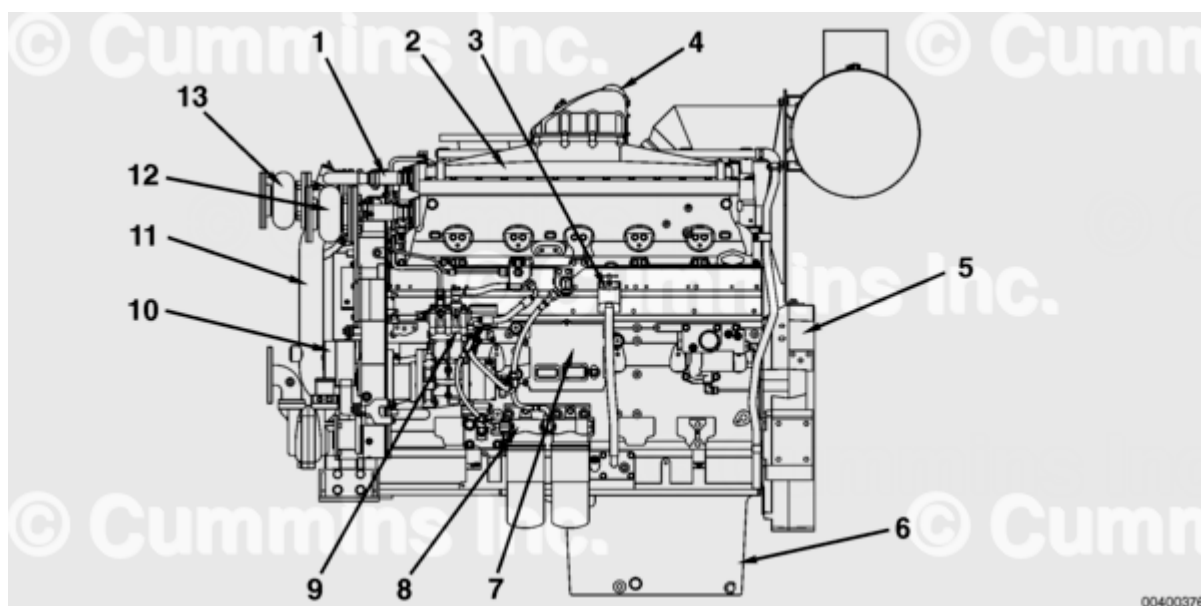
1. Fuel injector supply lines
2. Injectors
3. Exhaust outlet
4. Turbocharger
5. Air crossover pipe
6. Air filter
7. Eliminator™*
8. Aftercooler
9. Heat exchanger
10. Raw water pump
11. Raw water pump inlet
12. Coolant vents
13. Expansion tank pressure cap
14. Expansion tank
15. Thermostat housing
16. Raw water outlet from heat exchanger.

* The Eliminator™ is an available option on QSK19 Modular Common Rail System Marine engines.



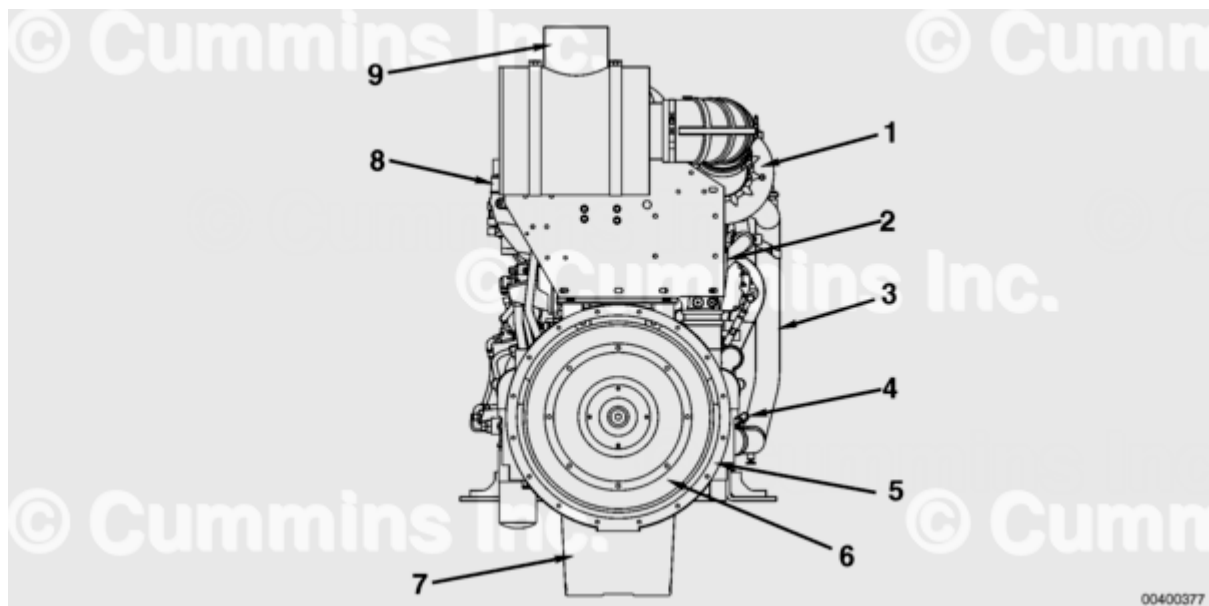
Front View - QSK19 CM850 Keel Cooled

1. Turbocharger
2. Exhaust outlet
3. Coolant from keel cooler
4. Air crossover pipe
5. Air filter
6. Aftercooler
7. Coolant to keel cooler
8. Lubricating oil pan
9. Vibration damper guard
10. Engine water pump inlet pipe
11. Alternator belt guard
12. Exhaust elbow.



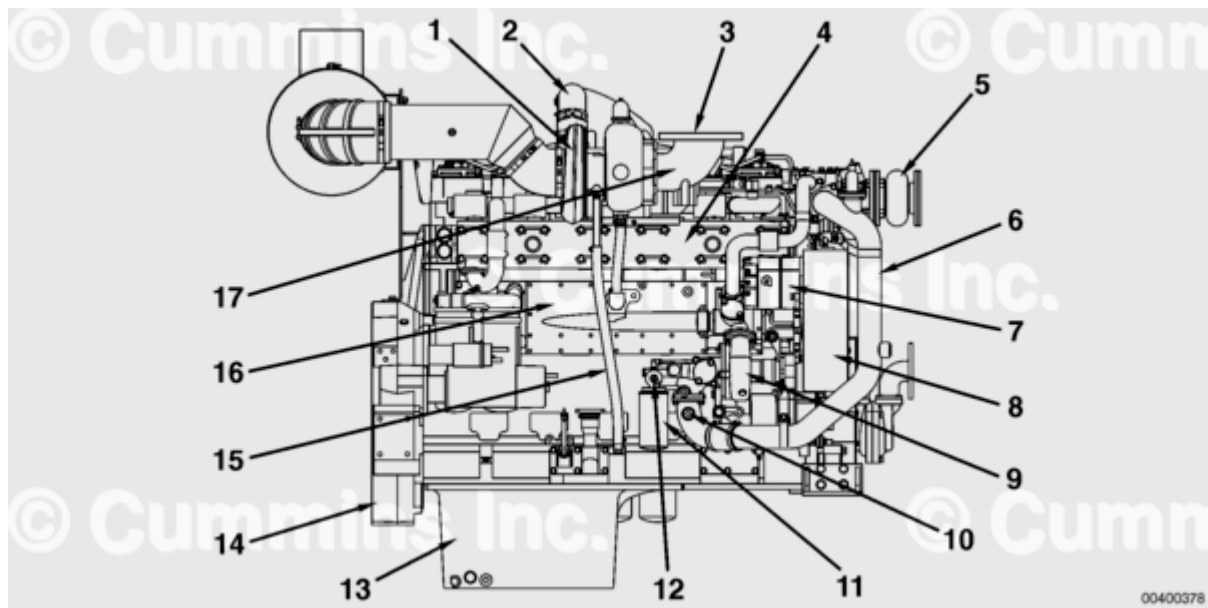
Left View - QSK19 CM850 Keel Cooled

1. Aftercooler coolant inlet
2. Aftercooler
3. Lubricating oil fill location
4. Air crossover pipe
5. Flywheel housing
6. Lubricating oil pan
7. ECM
8. Fuel filter head
9. Fuel pump
10. Vibration dampers
11. Engine water pump inlet pipe
12. Coolant to keel cooler
13. Coolant from keel cooler.



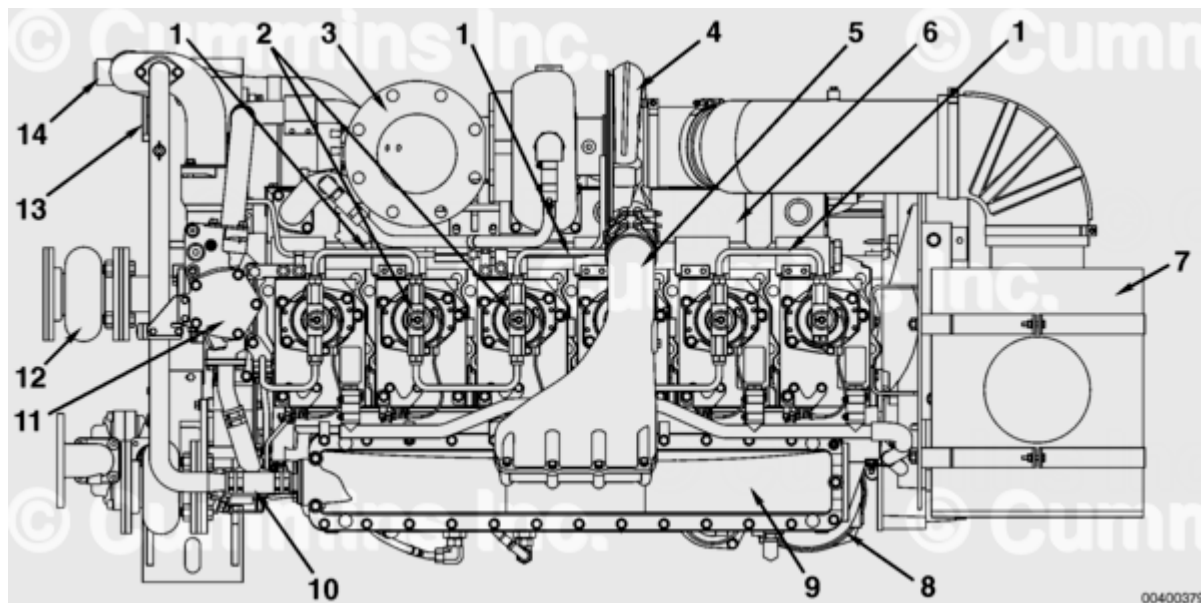
Rear View - QSK19 CM850 Keel Cooled

1. Turbocharger
2. Turbocharger oil drain
3. Engine water pump inlet pipe
4. Lubricating oil dipstick
5. Flywheel housing
6. Flywheel
7. Lubricating oil pan
8. Aftercooler
9. Air inlet.



Right View - QSK19 CM850 Keel Cooled

1. Turbocharger
2. Air crossover pipe
3. Exhaust outlet
4. Exhaust manifold
5. Coolant from keel cooler
6. Engine water pump inlet pipe
7. Alternator
8. Alternator belt guard
9. Water pump
10. Water inlet restriction test port
11. Water filter
12. Water filter shutoff valve
13. Lubricating oil pan
14. Flywheel housing
15. Turbocharger oil drain
16. Lubricating oil cooler
17. Exhaust elbow.



Top View - QSK19 CM850 Keel Cooled

1. Fuel Injector supply lines
2. Injectors
3. Exhaust outlet
4. Turbocharger
5. Air crossover pipe
6. Exhaust manifold
7. Air filter
8. Eliminator™*
9. Aftercooler
10. Aftercooler coolant inlet
11. Thermostat housing
12. Coolant to keel cooler
13. Alternator belt guard
14. Engine water pump inlet pipe.

* The Eliminator™ is an available option on QSK19 Modular Common Rail System Marine engines.